

Data Engineering Overview

The Infrastructure Behind Every Data Product

■■ FOUNDATIONS

■ LEARNING OUTCOME

By the end of this module you will understand exactly what data engineers build, how DE fits into the modern data team, the complete modern data stack, and a clear learning roadmap to your first DE role.

■ Skills You Will Gain

- Define data engineering and distinguish it from data science and analytics
- Describe the modern data stack — ingestion, storage, transform, serve
- Distinguish batch vs micro-batch vs streaming pipelines
- Map the core DE tools by category (ingestion, compute, orchestration, warehouse)
- Understand data quality principles and the DE skill roadmap
- Read and explain a real-world DE architecture diagram

■ Data Engineers earn ■10–45 LPA in India and \$100K–\$200K globally. Every data scientist, analyst, and ML engineer depends on the infrastructure DE builds.

SECTION 1

What is Data Engineering?

Data Engineering is the discipline of designing, building, and operating the **infrastructure, pipelines, and systems** that collect, store, process, and serve data reliably at scale — so that analysts, scientists, and applications can use it.

Data Engineering vs Data Science vs Analytics

Data Engineer: Builds the pipes, tanks, and pumps that move and store data
Data Analyst: Reads the data to answer business questions (SQL, dashboards)
Data Scientist: Builds statistical models and ML on top of prepared data
ML Engineer: Deploys ML models to production and keeps them running
Key insight: None of the other roles can work without reliable data infrastructure.

SECTION 2

The Modern Data Stack

1	Ingest Extract data from sources: databases, APIs, files, events — tools: Fivetran, Airbyte, custom scripts
2	Store Raw Land raw data in a data lake: S3, GCS, Azure Blob — cheap, durable, any format, schema-on-read
3	Process Transform raw data with Spark, dbt, or Glue — clean, validate, enrich, aggregate
4	Warehouse Load structured, analysis-ready data into: Snowflake, BigQuery, Redshift — fast SQL queries
5	Serve Expose data to consumers: BI dashboards (Looker, Tableau), APIs, ML feature stores
6	Orchestrate Schedule and monitor all pipeline tasks: Airflow, Prefect, Dagster — retry on failure
7	Observe Monitor data quality, freshness, volume, and schema changes: Monte Carlo, Great Expectations

SECTION 3

Batch vs Streaming Pipelines

Pipeline Type	Description	Latency	Tools
Full Batch	Process ALL data on a schedule	Hours–Daily	Spark, dbt, Airflow
Incremental	Process only new/changed rows since last	Minutes–Hourly	Spark, dbt (incremental)
Micro-batch	Process small windows every few seconds	Seconds	Spark Structured Streaming
Real-time Stream	Process events one-by-one as they arrive	Milliseconds	Kafka, Flink, Kinesis

Core DE Tools by Category

Category	Tools & Technologies
Ingestion / EL	Fivetran (managed connectors), Airbyte (open source), custom Python scripts, Debezium (C)
Data Lake Storage	AWS S3, Google Cloud Storage (GCS), Azure Data Lake Storage Gen2 (ADLS)
Processing	Apache Spark (PySpark), Apache Flink, dbt (SQL transforms inside warehouse)
Data Warehouse	Snowflake, Google BigQuery, Amazon Redshift, Azure Synapse Analytics
Streaming	Apache Kafka, Amazon Kinesis, Google Pub/Sub, Apache Flink
Orchestration	Apache Airflow, Prefect, Dagster, AWS Step Functions, dbt Cloud scheduler
Data Quality	Great Expectations, dbt tests, Soda, Monte Carlo, Datafold
File Formats	Parquet (columnar, compressed), Delta Lake (ACID), Iceberg, ORC, Avro

ETL vs ELT — The Paradigm Shift

ETL (old) vs ELT (modern)

ETL — Extract, Transform, Load: Extract from source → Transform in a separate ETL server → Load clean data into warehouse Problem: separate compute, slow, rigid schemas, expensive to scale ELT — Extract, Load, Transform: Extract from source → Load raw data into warehouse/lake → Transform using warehouse's own compute Why better: cloud warehouses (Snowflake, BigQuery) have unlimited scalable compute Transform with SQL (dbt) — version controlled, testable, collaborative

Data Engineer Skill Roadmap

Phase	Skills to Master
Phase 1 — Foundation	SQL (advanced), Python (pandas, requests), Linux CLI, Git version control
Phase 2 — Core DE	Databases (PostgreSQL, NoSQL), cloud storage (S3/GCS), Airflow orchestration
Phase 3 — Scale	Apache Spark (PySpark), Kafka streaming, dbt data modelling
Phase 4 — Cloud	AWS/GCP/Azure data services, infrastructure as code (Terraform)
Phase 5 — Advanced	Data quality frameworks, data contracts, real-time architectures, MLOps

■ PRACTICE Q & A

Q1. What is the difference between a data engineer and a data scientist?

A: Data engineers build and maintain the infrastructure, pipelines, and data systems. Data scientists use that clean, reliable data to build statistical models and ML. DE is infrastructure; DS is application.

Q2. What is the difference between a data lake and a data warehouse?

A: Data lake: raw, unstructured or semi-structured data stored cheaply in object storage (S3, GCS). No fixed schema. Data warehouse: cleaned, structured, optimised for SQL analytics (Snowflake, BigQuery). Fast queries.

Q3. What is ELT and why has it replaced ETL in modern data stacks?

A: ELT: Extract raw data, Load into warehouse, Transform with warehouse compute. Replaced ETL because cloud warehouses (Snowflake, BigQuery) have unlimited scalable compute. No separate ETL server needed.

Q4. Name 3 data pipeline orchestration tools and what they do.

A: Airflow: DAG-based scheduler, widely adopted. Prefect: Python-native, better error handling. Dagster: asset-based orchestration with built-in observability. All schedule, monitor, and retry pipeline tasks.

Q5. What is the Medallion architecture in data engineering?

A: A layered approach: Bronze (raw ingested data, exact copy of source) → Silver (cleaned, validated, deduplicated) → Gold (aggregated, business-ready, mart tables). Each layer adds quality and structure.

■■ DE Overview Cheat Sheet	
ETL	Extract → Transform → Load (legacy)
ELT	Extract → Load → Transform (modern)
Data Lake	Raw storage – S3, GCS, ADLS
Data Warehouse	Structured analytics – Snowflake, BigQuery
dbt	SQL transforms with tests and version control
Airflow	Pipeline orchestration – DAG-based
Kafka	Real-time event streaming
Spark	Distributed large-scale data processing
Fivetran / Airbyte	Automated data ingestion connectors
Batch Pipeline	Scheduled – hourly or daily processing
Streaming Pipeline	Continuous event-by-event processing
Medallion Pattern	Bronze → Silver → Gold data layers